

Part: **3.4 – STRENGTH OF COMPOSITE PARTS**

Chapter: **3450**

Title: **BUCKLING OF COMPOSITE PLATES**

Revision: **1.0**

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1 INTRODUCTION

1.1 OBJECT

This chapter describes the methodology proposed for calculating the buckling onset of rectangular flat plates made of composite laminates. Two sets of methods are presented:

✓ Pre-sizing methods

To be used in initial phases of the design process. They are based in exact solutions or approximations found in literature, engineering handbooks, or in internal Airbus DS development in some cases.

✓ Methods for final strength check

To be used in final sizing or in certification strength justification. They are based in numerical solutions, have a higher computational cost and shall be utilized only using validated software implementations.

1.2 SCOPE AND LIMITATIONS

Buckling of composite plates. Limitation of pre-sizing methods:

- ✓ Rectangular shape
- ✓ Flat plate (simply) supported on all its perimeter
- ✓ Other limitations are listed in the dedicated sub-chapter

In any case, the post-buckling behavior shall not be discussed here.

2 METHODS FOR PRE-SIZING

The pre-sizing methods consist of analytical solutions, handbook methods and available COTS software (such as ESDU programs) with specific simplifications and resulting advantages and disadvantages. The following sub-chapters present some available pre-sizing methods for three distinct use cases: 1. specially orthotropic plates, 2. plates with bending-twisting coupling, 3. specially orthotropic plates including transverse shear stiffness

2.1 SPECIALLY ORTHOTROPIC PLATES

Applicability of the following formulae is strictly limited to specially orthotropic laminates, i.e. those symmetric ($B_{ij}=0$) and for which the bending-twisting coupling terms vanish ($D_{16}=D_{26}=0$). The last condition is always true for symmetric laminates made of fabric materials ($E_1=E_2$). In practice, for symmetric laminates made of tape materials, when the stacking sequence follows the standard design rules (plies balanced, 45° plies in pairs $+45/-45$, ...), the expressions below will be also a reasonably good approximation. However, one must be careful about this point as it will lower the critical buckling load.

The CMH-17 (Ref. 2) provides a formula to calculate a “corrected” equivalent or generalized bending stiffness if the laminate is only slightly unsymmetric/unbalanced. This results in lower bending stiffness values in the D matrix and thus lowers critical buckling loads. However the buckling mode or other properties cannot be described with this correction.

$$[D^*] = [D] - [B][A]^{-1}[B] \quad \text{see for example CMH-17, Vol. 3, formula 9.2.2.1(b)}$$

Additionally, the effect of the through-thickness (or transverse) shear stiffness in buckling stability is neglected in these formulations. The shear stiffness is assumed infinite and the stability driven by bending, what it is a good approximation for thin walled parts ($t \ll b$).

Buckling under biaxial loads

A closed form solution of the buckling equation can be found for the case of a simply supported, specially orthotropic plate ($D_{16} = D_{26} = 0$; e.g. cross-ply laminates) under bi-axial loading, (e.g in Lekhnistkii's book, equation 101.3).

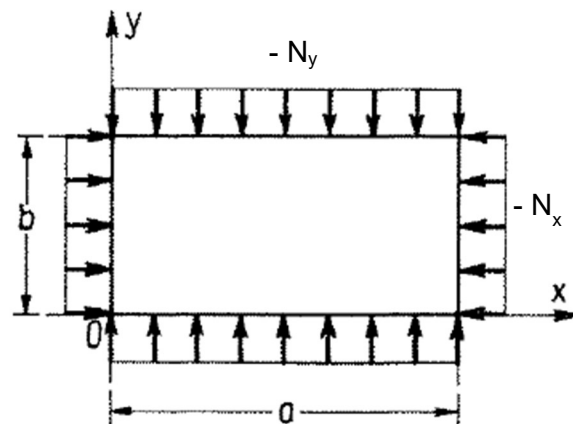


Figure 1. Simply supported rectangular plate under bi-axial compression

Adapted to current notation and signs criteria results:

$$N_{xb} \cdot (m/a)^2 + N_{yb} \cdot (n/b)^2 = - \pi^2 \cdot [D_{11} \cdot (m/a)^4 + 2 \cdot (D_{12} + 2 \cdot D_{66}) \cdot (m \cdot n)^2 / (a \cdot b)^2 + D_{22} \cdot (n/b)^4]$$

... where:

- a is the length of the plate
- b is the width of the plate
- m is the number of buckling semi-waves in longitudinal direction
- n is the number of buckling semi-waves in transversal direction
- N_{xb}, N_{yb} are the loads at the buckling onset (negative values if compression)
- $D_{11}, D_{12}, D_{22}, D_{66}$ are the non-zero terms of the flexural stiffness matrix of the laminate

If buckling load is expressed in terms of times the applied loads, i.e. the Buckling Level (equivalent to Reserve Factor if buckling were considered failure)...

$$N_{xb} = BL \cdot N_x, \quad N_{yb} = BL \cdot N_y$$

... the solution becomes:

$$BL = - \pi^2 \cdot [D_{11} \cdot (m/a)^4 + 2 \cdot (D_{12} + 2 \cdot D_{66}) \cdot (m \cdot n)^2 / (a \cdot b)^2 + D_{22} \cdot (n/b)^4] / [N_x \cdot (m/a)^2 + N_y \cdot (n/b)^2]$$

The critical buckling load solution shall be found by iterating over the possible number of semi-waves (m, n from 1 to ...) to find the minimum value. Typically the minimum is found for one semi-wave in the shortest side direction ($n = 1$ if $a > b$) and the ratio m/n of the order of a/b .

Buckling under shear loads

Analogous to the bi-axial loading, a solution for buckling under pure shear can be found in literature. A graphical handbook solution based on the results of Seydel and Davenport and Bert is presented in the HSB 45112-02. Note that α is the effective aspect ratio of the plate and β is Seydel's orthotropy parameter:

$$\alpha = b/a \cdot (D_{11} / D_{22})^{1/4}$$

$$\beta = (D_{12} + 2 \cdot D_{66}) / (D_{11} \cdot D_{22})^{1/2}$$

With the buckling factor k obtained from the following figure, the critical buckling in-plane section shear force q_{cr} per unit width is calculated as:

$$q_{cr} = k \cdot (\pi / b)^2 \cdot (D_{11} \cdot D_{22}^3)^{1/4}$$

The in-plane mean shear stress can be obtained with: $\tau_{cr} = q_{cr}/t$

For the frequent case of high aspect ratios, the formula of a plate with $\alpha = 0$ (infinite plate) is given below (small conservative error for $a/b \geq 3$):

$$\beta > 1: \tau_{cr} = 4 / (t \cdot b^2) \cdot (D_{22} \cdot (D_{12} + 2 \cdot D_{66}))^{1/2} \cdot (11.7 + 0.532 / \beta + 0.938 / \beta^2)$$

$$\beta \leq 1: \tau_{cr} = 4 / (t \cdot b^2) \cdot (D_{11} \cdot D_{22}^3)^{1/4} \cdot (8.12 + 5.05 \cdot \beta)$$

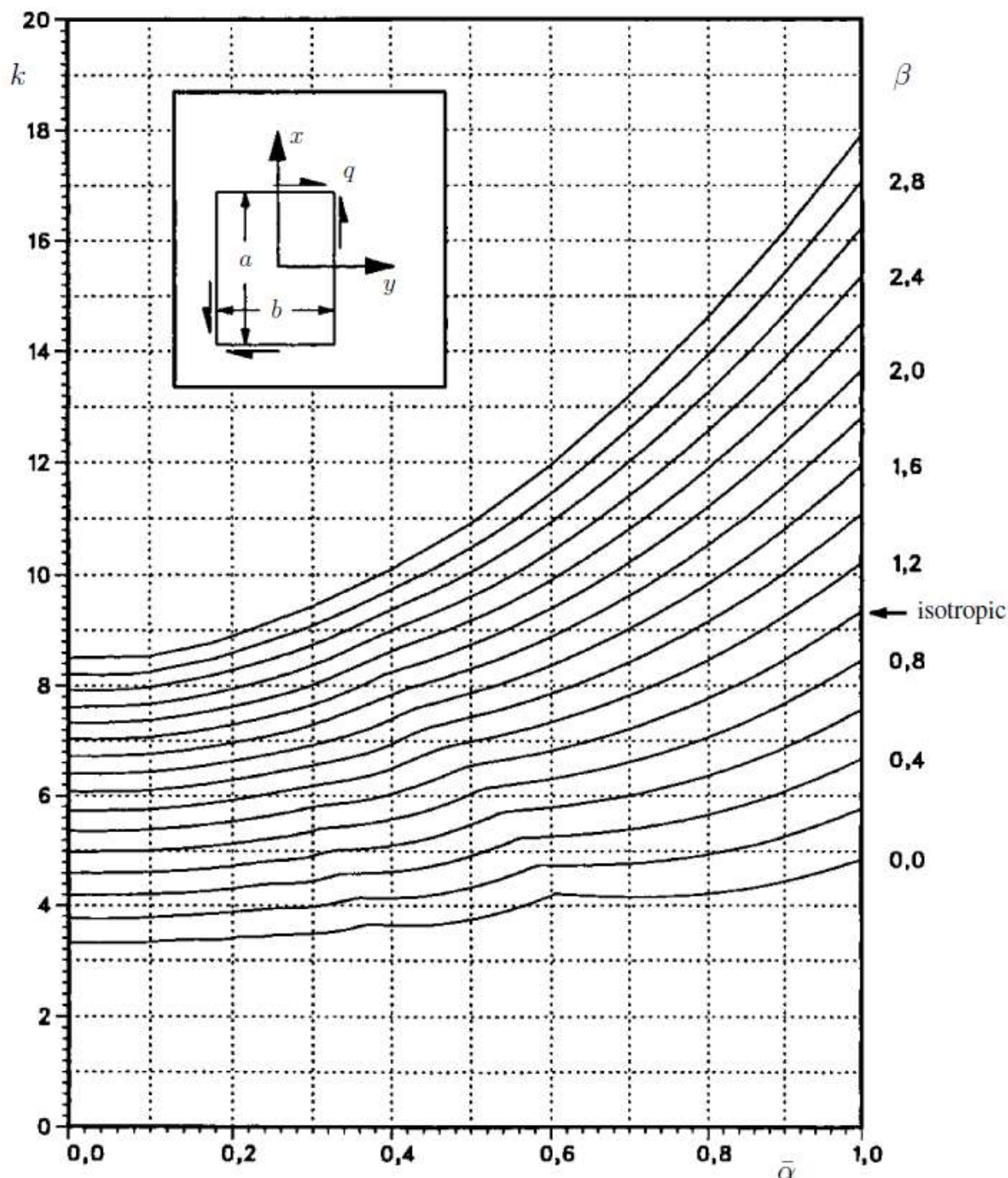


Figure 2. Figure 1 in HSB 45112-02

Buckling under linearly distributed loads

For the case of buckling under linearly distributed loading, a graphical handbook solution based on the results of Lekhnitskii is presented in HSB 45111-09. The parameters α and β are defined in the same way as above. For the special case of a pure in-plane bending, one can directly use the

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following equations from Lekhnitskii's book by iterating over the half wave m in order to obtain the critical buckling moment:

$$M_{cr} = \frac{\pi^2 \sqrt{D_1 D_2}}{6} k$$

With:

$$k = \frac{9\pi^2}{32} \sqrt{a_{m1} a_{m2}}$$

$$a_{m1} = \sqrt{\frac{D_{11}}{D_{22}} \left(\frac{m}{c}\right)^2} + \frac{2(D_{12} + 2D_{66})}{\sqrt{D_{11} D_{22}}} + \sqrt{\frac{D_{22}}{D_{11}} \left(\frac{c}{m}\right)^2}$$

$$a_{m2} = \sqrt{\frac{D_{11}}{D_{22}} \left(\frac{m}{c}\right)^2} + \frac{8(D_{12} + 2D_{66})}{\sqrt{D_{11} D_{22}}} + 16 \sqrt{\frac{D_{22}}{D_{11}} \left(\frac{c}{m}\right)^2}$$

$$c = \frac{a}{b}$$

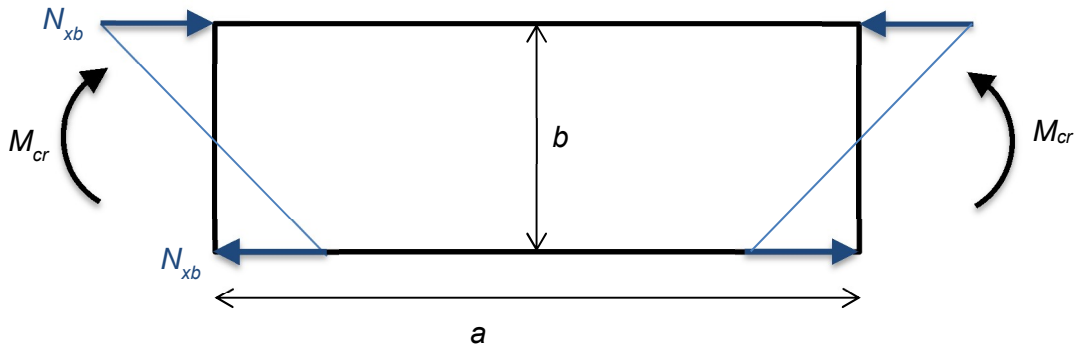


Figure 3. Rectangular plate under in-plane bending

The maximum compression and tension axial loads at plate edges, at buckling onset under pure in-plane bending, assuming a linear load distribution, would be then:

$$N_{xb} = 6 \cdot M_{cr} / b^2$$

The minimum value of k can be found for $c = \frac{m}{\sqrt{2}} \sqrt{D_{11}/D_{22}}$ and thus for the infinite plate $c \rightarrow \infty$ with:

$$k = 11.1 \cdot \left(1.25 + \frac{D_{12} + 2D_{66}}{\sqrt{D_{11} D_{22}}} \right)$$

Warning! The formula above may lead to slightly unconservative predictions (5% approx.) when compared with more accurate methods. Typically the HSB 45111-09 solution (or equivalent numerical methods) shall be preferred.

2.2 LONG PLATES WITH BENDING-TWISTING COUPLING

The previous shown equations are not valid for laminates possessing a coupling between bending-twisting, i. e. $D_{16} \neq 0$ and $D_{26} \neq 0$. With increasing values of D_{16} and D_{26} , these buckling formulae become unconservative for compression and either conservative or unconservative for shear depending on the load direction. However, in certain applications, e.g. aerodynamic tailoring of a wing, the usage of laminates with substantial values of D_{16} and D_{26} is wanted and thus the stability of such structures has to be evaluated. The assumption of (infinitely) long plates and usage of non-dimensional parameters allows deriving approximate closed form solutions for symmetric laminates with bending-twisting coupling as presented by Weaver. Aspect ratios of $a/b \geq 3$ are expected to satisfy the long plate assumption. Furthermore transverse shear stiffness is said to be infinite ($t \ll b$).

Buckling under biaxial loads

Weaver proposed a solution for the buckling problem of long simply supported plates made of an anisotropic laminate subjected to bi-axial compression loads (*), in Ref. 13. The implementation of that methodology is far from being trivial, and the formulation will not be reproduced here. The reader can read that reference for details.

A closed form solution for the case without transverse load is discussed in following paragraph.

(*) Tensile load in transverse direction ($N_y > 0$) combined with longitudinal compression ($N_x > 0$) are also covered by the formulation.

Buckling under longitudinal compression load

The same solution discussed in chapter 3430 "LOCAL BUCKLING AND CRIPPLING OF BEAMS/STRINGERS" can be reminded here. Herencia, Weaver and Friswell derived expressions for calculating the compression buckling load of long plates with flexural/twist anisotropy (Ref. 18):

$$N_{xb} = K_x \cdot \pi^2 \cdot (D_{11} \cdot D_{22})^{1/2} / b^2$$

Where the buckling coefficient K_x , for simply supported conditions, is:

$$K_x = 2 \cdot (1 - 4 \cdot \delta \cdot \gamma - 3 \cdot \delta^4 + 2 \cdot \delta^2 \cdot \beta)^{1/2} + 2 \cdot (\beta - 3 \cdot \delta^2)$$

...expressed in terms of the non-dimensional parameters suggested by Nemeth:

$$\begin{aligned} \alpha &= (D_{22}/D_{11})^{1/4} & \beta &= (D_{12} + 2 \cdot D_{66}) / (D_{11} \cdot D_{22})^{1/2} \\ \gamma &= D_{16} / (D_{11}^3 \cdot D_{22})^{1/4} & \delta &= D_{26} / (D_{11} \cdot D_{22}^3)^{1/4} \end{aligned}$$

Buckling under shear load

Using the same non-dimensional parameters as above, a closed form solution can be obtained for the case of uniform shear in infinitely long plates:

$$N_{xy} = K_{xy} \frac{\pi^2}{b^2} \sqrt{D_{11} D_{22}^3}$$

$$K_{xy} = 3.42 + 2.05\beta - 0.13\beta^2 - 1.79\gamma - 6.89\delta + 0.36\beta(2\gamma + \delta) - 0.25(2\gamma + \delta)^2$$

2.3 SPECIALLY ORTHOTROPIC PLATES WITH TRANSVERSE SHEAR STIFFNESS

Both previously presented pre-sizing methods assume infinite transverse shear stiffness and are therefore unsuitable when analysing moderately thick plates ($t/b \geq 0.05$). Therefore methods including transverse shear stiffness may be of interest for such plates, e.g. sandwich panels.

A closed form solution for this case is not known to the author. However, available COTS software based on energy methods such as ESDU 81047 provide results for such cases. This ESDU method cannot handle linearly distributed loads as it is the case for in-plane bending.

2.4 EFFECTIVE SUPPORT

Many structural components in the airframe include rectangular flat plates with slenderness ratios (a/b) significantly greater than 1 (e.g. ribs in lift boxes, with the chord dimension much greater than the box height). A typical method to retard the buckling onset in this kind of panels is to install transversal stiffeners to split the part in subpanels with a/b ratios around 1.

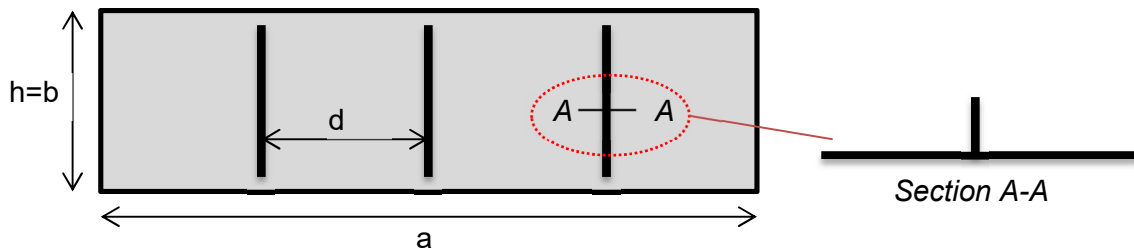


Figure 4. Shear web with vertical stiffeners to retard buckling onset

For shear resistant webs made of isotropic materials, Bruhn discusses, in his book, a proposal of H. Wagner for estimating the minimum moment of inertia required for the transverse stiffeners in order to prevent the global buckling. Later on, a more recent alternative is shown, preferred for not being dependent on the applied load (see Ref. 7, section C10.10).

$$I_{vst} / (d \cdot t^3) = 0.0217 / [d / (h \cdot K_s^{0.5})]^{8/3}$$

Bruhn Fig. C10.9

Where:

- I_{vst} Moment of inertia of the stiffener (without eff. skin)
- d Distance between stiffeners (d must be lower than h)
- h Height of the web plate
- t Web thickness
- K_s Shear buckling coefficient (that in formulas for critical buckling shear stress $F_{s,cr} = K_s \cdot E \cdot [t/b]^2$, with b the minimum between h and d , thus d)

However, an alternative to that criterion applicable to laminated composite plates is not really available. Consider the following an “extrapolated” extension of the proposal above, which should only be considered illustrative, a non-validated method.

$$EI_{vst} \cdot [D^{-1}]_{22} / d = 0.26 / [d / (h \cdot K_s^{0.5})]^{8/3}$$

Where:

EI_{vst} Bending stiffness of the stiffener cross section

$[D^{-1}]_{22}$ Term (2,2) of the inverted bending stiffness matrix of the web laminate (assumed the 0° material axis aligned with the longitudinal/long direction of the plate). Notice this value is associated with the plate bending modulus and thickness as: $1/[D^{-1}]_{22} = (E_{by} \cdot t^3)/12$

K_s Coefficient derived from the buckling shear load of the laminated plate N_{sb} of dimensions $d \times h$ and web thickness as: $K_s = N_{sb} / (E_{by} \cdot t^3/d^2)$

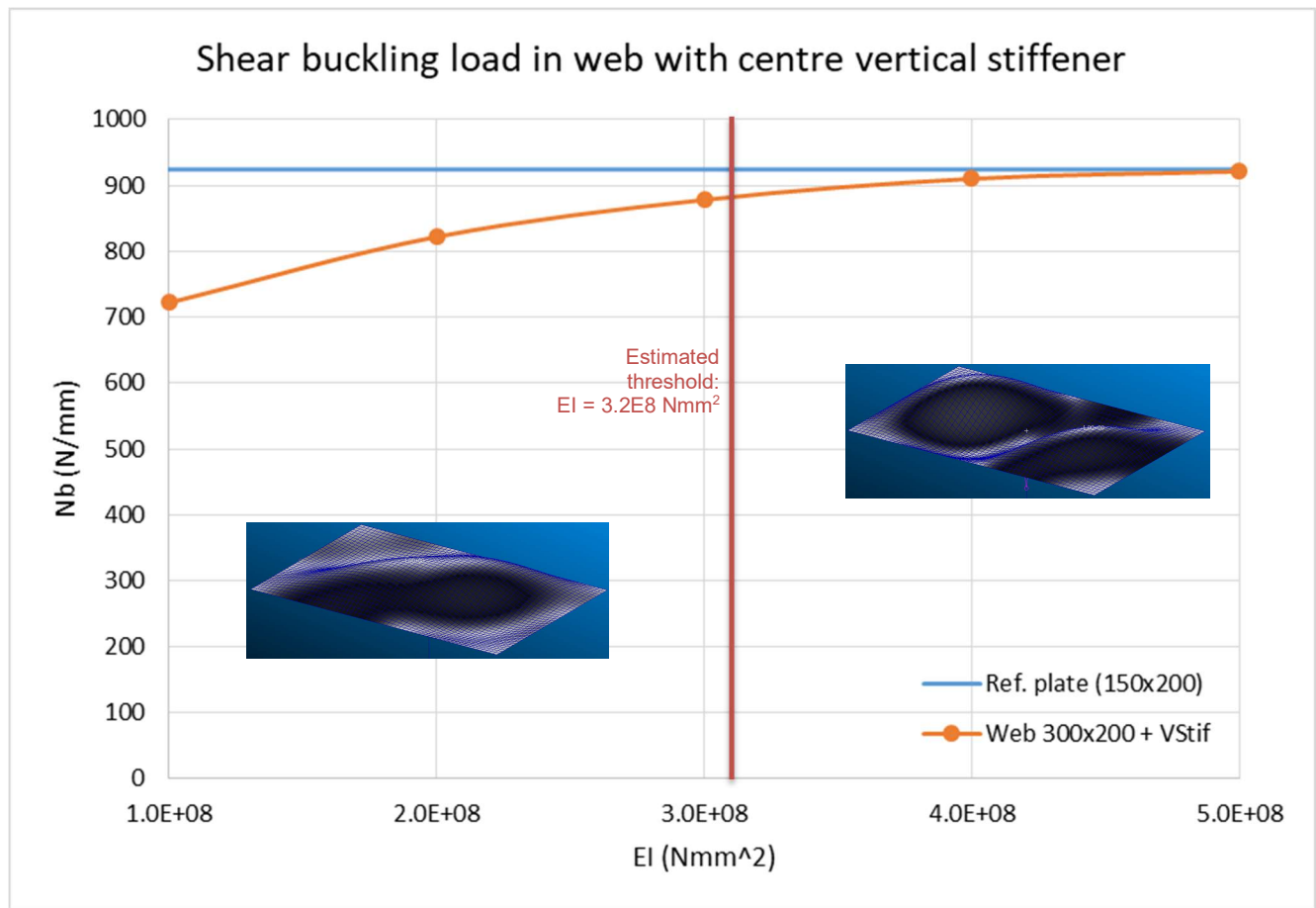


Figure 5. FEM shear buckling with central vertical stiffener of increasing bending stiffness

Remark: Previous discussion and expressions need future check and, probably, adjustments or fixings.

2.5 BUCKLING UNDER COMBINED LOADS

Abovementioned references present typical interaction formulas used to obtain the safety margin vs buckling stability in multi-axial stress states. The proposed failure law is:

$$R_c + R_f^{1.76} + R_s^2 = 1$$

...where R_c , R_f and R_s are the failure indices in uni-axial loads for axial longitudinal compression, in-plane bending (flexure) and shear respectively, i.e.:

$$R_c = 1 / BL \quad \dots \text{if obtained from the explicit solution available for bi-axial loads}$$

This one is the recommended approach, more precise, even less straightforward or fast than the following.

$$= R_x + R_y \quad \dots \text{from uni-axial components, with } R_x = -N_x / N_{xb} \text{ , } R_y = -N_y / N_{yb}$$

When any of the axial forces N_x or N_y is positive tension the recommended (conservative) approach is to make the corresponding failure index equal to 0 in the interaction formula.

$$R_f = M / M_b$$

$$R_s = N_{xy} / N_{xyb}$$

That expression can be expressed introducing the Reserve Factor for performing the strength check of a general load case:

$$RF \cdot R_c + (RF \cdot R_f)^{1.76} + (RF \cdot R_s)^2 = 1$$

If the in-plane applied bending M is not zero, the Reserve Factor shall have to be solved numerically.

Buckling of rectangular plates under (bi)axial and shear load

If the in-plane applied bending M is zero, and then the plate is subjected to uniform (bi)axial and shear loads, there is a closed form solution for the Reserve Factor in equation above:

$$RF = 2 / [R_c + (R_c^2 + 4 \cdot R_s^2)^{0.5}] \quad (\text{Bruhn eq. C5.12})$$

3 METHODS FOR FINAL STRENGTH CHECK

The methods presented in previous section are generally, conservative, and indeed may still be used in final sizing steps, or in certification justification, provided their limitations have not been exceeded. In addition to those inherent limitations of the analytical methods, under some circumstances, more refined methodologies shall be needed to evaluate the accurately the buckling levels of the part. These circumstances may include short static margins, weight saving campaigns, loads exceedances in final load loops, ..., among others.

At Airbus DS engineering office, the available implementations of these more refined tools (all supported in numerical methods) include:

Energy (Rayleigh-Ritz) methods

These methods compute the total energy of the structural system as a sum of the deformation energy of the structure plus the potential energy of the applied loads. The deformed shape is approximated by series of functions compatible with the boundary conditions, with the coefficients of these functions unknown. The solution for the coefficients is obtained deriving the total energy and making it equal to zero for searching its minimum. The eigenvalues and eigenvectors of the resulting linear system of equations are the buckling levels and buckling modes.

Current implementations cover rectangular plane plates, un-stiffened or stiffened with horizontal or vertical stringers. They are capable to detect local and global buckling modes, and to model rotational constraints at supports (beyond the simple-supports conditions).

Details in Ref. 9, Ref. 15, Ref. 16 (Airbus DS engineering reports) or Ref. 17.

Detailed Finite Elements Models

For specially complicated parts the use of FEM analysis is the last and more sophisticated tool to solve the buckling analysis problem.

Rules and recommendations for adequate preparation and processing FEM analyses are out of the scope of this chapter. See Volume 2 of the Stress Methods Manual for further information.

4 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
a	mm	Length of the plate/shell (dimension in X-longitudinal direction)
$[A]$	N/mm	In-plane stiffness matrix of a composite laminate
b	mm / -	Width of the plate/shell (dimension in Y-transversal direction) or, as subscript, relative to buckling
c	-	As subscript, relative to compression
CRFP	-	Carbon Fiber Reinforced Polymer
COTS	-	Commercial off-the-shelf
$[B]$	N	Matrix defining the laminate shear-bending coupling
BL	-	Buckling level, ratio between buckling load and applied load
$[D]$	Nmm	Flexural stiffness matrix of a composite laminate
f	-	As subscript, relative to in-plane bending
I	mm ⁴	Bending moment of inertia
M_b	Nmm	In-plane bending moment of the plate at buckling onset
N_{xb}	N/mm	Compression buckling load-flow (force per unit width) in X direction
N_{xyb}	N/mm	Shear buckling load-flow (force per unit width)
N_{yb}	N/mm	Compression buckling load-flow (force per unit width) in Y direction
q	N/mm	In plane shear force per unit width
R	mm / -	Curvature radius of the shell (around an axis parallel to X direction) or, form context, failure index, i.e. ratio between uni-axial load and allowable load
RF	-	Reserve Factor, ratio between allowable load and applied load
s / xy	-	As subscripts, relative to in-plane shear
t	mm	Thickness of the plate
τ	N/mm ²	In-plane mean shear stress
UD	-	Unidirectional

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REFERENCES

<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1 [S. G. Lekhnitskii]	1984	Anisotropic Plates
Ref. 2 [SAE] CMH-17	2012	Composite Materials Handbook (Volume 3)
Ref. 3 ESDU80023	1995	BUCKLING OF RECTANGULAR SPECIALLY ORTHOTROPIC PLATES
Ref. 5 ESDU 81047	-	BUCKLING OF FLAT RECTANGULAR PLATES
Ref. 7 [E. F. Bruhn]	1973	ANALYSIS AND DESIGN OF FLIGHT VEHICLE STRUCTURES
Ref. 8 [Michael C. Y. Niu]	2 nd Ed.	AIRFRAME STRESS ANALYSIS AND SIZING
Ref. 9 NT-T-ADP-06002	Ed. A	Manual teórico del programa PANPA, versión 5.0
Ref. 10 HSB 45112-02	B	Shear buckling of flat, rectangular, orthotropic plates
Ref. 11 HSB 45111-09	B	Buckling of unstiffened, orthotropic, flat, rectangular plates under linearly distributed compressive loading
Ref. 12 [Paul M. Weaver]	2004	On Optimisation of Long Anisotropic Flat Plates subject to Shear Buckling Loads
Ref. 13 [Paul M. Weaver]	2005	Design Formulae for Buckling of Biaxially Loaded Laminated Rectangular Plate with Flexural/Twist Anisotropy
Ref. 14 [Paul M. Weaver]	2006	Approximate analysis for buckling of compression loaded long rectangular plates with flexural/twist anisotropy
Ref. 15 TEA-T-NT-210048	Ed. B	Manual Teórico del Programa ARPA
Ref. 16 NT-T-ADP-00005	Ed. A	MANUAL TEORICO DEL PROGRAMA PANSA, Versión 4.0
Ref. 17 [J. N. Reddy]	2 nd Ed.	MECHANICS of LAMINATED COMPOSITE PLATES and SHELLS. Theory and Analysis.
Ref. 18 [Herencia, Weaver, Friswell]	2010	Closed-Form Solutions for Buckling of Long Anisotropic Plates with Various Conditions under Axial Compression

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SOFTWARE

Program	Version	Description
STRENGTH2000 (Airbus DS)	TBC	Detailed sizing of metallic and composite aircraft structures
NASTRAN (MSC-Software)	2005	Multidisciplinary structural analysis software based in the Finite Elements method
PANPA, SAM (CFSKIN), SPAEM (Airbus DS)	5.4 0.2 0.9	Buckling of rectangular panels (energy Rayleigh-Ritz method)

Table 1. Software programs referenced in this chapter

RECORD OF REVISIONS

Revision Date	Authors	Change Reason	Pages/Sub-chapters affected
1.0 27/10/2021	G. Baños F. Glock	First issue	All pages

Table 2. Record of Revisions: authors, change reasons and sections/pages affected

APPROVAL SIGNATURES TABLE

Dpt. \ Site	Getafe	Manching
Stress Methods (TEPAP-TL4)	G. Baños 27/10/2021	Florian Glock 27/10/2021

Table 3. Approval signatures table for last revision of this chapter